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"""
CONN Capacity Test: Scaling to N=1024
Parameters:  $\lambda=0.5$  (annealed),  $\eta=0.02$ , steps=400, jitter=0.5, noise=0.30
"""

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import numpy as np

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def generate_patterns(M, N, seed):
    np.random.seed(seed)
    return np.random.choice([0, np.pi], size=(M, N))

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def bitflip_plus_jitter(pattern, noise_level, jitter_std, rng):
    noisy = pattern.copy()
    mask = rng.random(len(pattern)) < noise_level
    noisy[mask] = (noisy[mask] + np.pi) % (2 * np.pi)
    noisy = noisy + rng.normal(0, jitter_std, len(pattern))
    return np.mod(noisy, 2 * np.pi)

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def compute_overlap(retrieved, target):
    r = np.where(np.cos(retrieved) >= 0, 0.0, np.pi)
    t = np.where(np.cos(target) >= 0, 0.0, np.pi)
    return np.sum(np.abs(np.cos(r) - np.cos(t)) < 0.5) / len(target) *
100

```

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def hebbian_weights(patterns):
    xi = np.cos(patterns)
    J = (xi.T @ xi) / patterns.shape[1]
    np.fill_diagonal(J, 0)
    return J

```

```

def hopfield_recall(J, noisy, max_steps=200):
    s = np.cos(noisy).copy()
    for _ in range(max_steps):
        s_new = np.sign(J @ s)
        s_new[s_new == 0] = 1
        if np.array_equal(s_new, s):
            break
        s = s_new
    return np.where(s > 0, 0.0, np.pi)

```

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def conn_annealed(J, noisy, lam, eta_phi, steps):
    N = len(noisy)
    phi = noisy.copy()
    A = np.ones(N)
    for step in range(steps):
        t = step / steps
        lam_t = lam * t
        s, c = np.sin(phi), np.cos(phi)
        coupling = (J @ (A * s)) * c - (J @ (A * c)) * s
        coherence = -2 * lam_t * A**2 * s * c

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    dphi = A * coupling + coherence
    phi = (phi + eta_phi * dphi) % (2 * np.pi)
    dA = -2 * lam_t * A * s**2 - 1.0 * (A - 1)
    A = np.clip(A + 0.03 * dA, 0.01, 2.0)
    return phi

#
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# MAIN EXPERIMENT - N=1024
#
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print("=" * 90)
print("CONN vs Hopfield Capacity Test - Ultra Scaling (N=1024)")
print("Parameters:  $\lambda=0.5$  (annealed),  $\eta=0.02$ , steps=400, jitter=0.5,
noise=0.30")
print("=" * 90)

noise = 0.30
jitter = 0.5
trials = 30
test_patterns_per_trial = 3

for N in [1024]:
    print(f"\n{'='*80}")
    print(f"                                N = {N}")
    print(f"{'='*80}")
    print(f"  {'M':>4s} {'Hopfield':>10s} {'CONN':>8s} {' $\Delta$ ':>8s}
{'hop_std':>8s} {'conn_std':>8s}")
    print(f"  {'-'*62}")

    hop_cap = 0
    conn_cap = 0
    max_M = int(0.35 * N)

    for M in range(1, max_M + 1):
        hop_ov = []
        conn_ov = []

        # Note: Pre-generating patterns for the trial to speed up
compute
        for trial in range(trials):
            patterns = generate_patterns(M, N, seed=trial)
            J = hebbian_weights(patterns)

            for mu in range(min(M, test_patterns_per_trial)):
                rng = np.random.RandomState(trial * 1000 + mu)
                noisy = bitflip_plus_jitter(patterns[mu], noise,
jitter, rng)

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        # Baseline
        ret_h = hopfield_recall(J, noisy)
        hop_ov.append(compute_overlap(ret_h, patterns[mu]))

        # CONN Recall
        ret_c = conn_annealed(J, noisy, 0.5, 0.02, 400)
        conn_ov.append(compute_overlap(ret_c, patterns[mu]))

    h_mean = np.mean(hop_ov)
    c_mean = np.mean(conn_ov)
    h_std = np.std(hop_ov)
    c_std = np.std(conn_ov)

    if h_mean >= 80: hop_cap = M
    if c_mean >= 80: conn_cap = M

    # Strategic printing for log readability
    if (M <= 5 or M % 10 == 0 or
        abs(h_mean - 80) < 5 or abs(c_mean - 80) < 5):
        print(f"    {M:4d} {h_mean:9.1f}% {c_mean:7.1f}% {c_mean-
h_mean:+7.1f}% {h_std:7.1f}% {c_std:7.1f}%")

    # Early stopping logic
    if h_mean < 55 and c_mean < 55 and M > 50:
        break

    ratio = conn_cap / hop_cap if hop_cap > 0 else 0
    print(f"\n    _____")
    print(f"    | Hopfield capacity: M={hop_cap:3d} ( $\alpha$ ={{hop_cap/N:.3f}})
|")
    print(f"    | CONN capacity:      M={{conn_cap:3d}} ( $\alpha$ ={{conn_cap/N:.3f}})
|")
    print(f"    | Improvement ratio: {ratio:.2f}×
|")
    print(f"    _____")

print("\n" + "="*90)
print("Scaling run completed.")

```

# RESULTS OF THE RUN N= 1024

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=====
CONN vs Hopfield Capacity Test - Ultra Scaling (N=1024)
Parameters:  $\lambda=0.5$  (annealed),  $\eta=0.02$ , steps=400, jitter=0.5, noise=0.30
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=====
=====
N = 1024
=====
=====
```

| M     | Hopfield | CONN  | $\Delta$ | hop_std | conn_std |
|-------|----------|-------|----------|---------|----------|
| ----- |          |       |          |         |          |
| 1     | 100.0%   | 90.6% | -9.4%    | 0.0%    | 3.2%     |
| 2     | 100.0%   | 91.0% | -9.0%    | 0.0%    | 3.0%     |
| 3     | 100.0%   | 91.4% | -8.6%    | 0.0%    | 2.5%     |
| 4     | 100.0%   | 91.6% | -8.4%    | 0.0%    | 2.6%     |
| 5     | 100.0%   | 91.7% | -8.3%    | 0.0%    | 2.3%     |
| 10    | 100.0%   | 92.4% | -7.6%    | 0.0%    | 2.2%     |
| 20    | 100.0%   | 92.1% | -7.9%    | 0.0%    | 2.1%     |
| 30    | 100.0%   | 92.0% | -8.0%    | 0.0%    | 2.1%     |
| 40    | 100.0%   | 91.6% | -8.4%    | 0.0%    | 2.1%     |
| 50    | 100.0%   | 90.8% | -9.2%    | 0.0%    | 2.3%     |
| 60    | 100.0%   | 90.3% | -9.7%    | 0.0%    | 2.5%     |
| 70    | 100.0%   | 89.5% | -10.5%   | 0.0%    | 2.5%     |
| 80    | 100.0%   | 88.9% | -11.1%   | 0.1%    | 2.5%     |
| 90    | 99.1%    | 88.1% | -11.1%   | 4.6%    | 2.5%     |
| 100   | 96.4%    | 87.2% | -9.3%    | 9.2%    | 2.8%     |
| 110   | 91.7%    | 86.3% | -5.3%    | 12.7%   | 2.8%     |
| 119   | 84.8%    | 85.5% | +0.6%    | 14.9%   | 2.8%     |
| 120   | 83.2%    | 85.4% | +2.2%    | 15.4%   | 2.8%     |
| 121   | 82.5%    | 85.3% | +2.8%    | 15.2%   | 2.8%     |
| 122   | 80.9%    | 85.3% | +4.4%    | 15.3%   | 2.7%     |
| 123   | 80.3%    | 85.2% | +4.9%    | 15.1%   | 2.8%     |
| 124   | 80.2%    | 85.1% | +4.9%    | 14.5%   | 2.8%     |
| 125   | 78.6%    | 85.0% | +6.4%    | 14.3%   | 2.8%     |
| 126   | 77.4%    | 84.8% | +7.5%    | 14.5%   | 2.8%     |
| 127   | 76.9%    | 84.8% | +7.9%    | 14.4%   | 2.8%     |
| 128   | 76.3%    | 84.8% | +8.4%    | 13.5%   | 2.7%     |
| 129   | 75.6%    | 84.7% | +9.1%    | 13.7%   | 2.8%     |
| 130   | 74.9%    | 84.7% | +9.8%    | 13.3%   | 2.7%     |
| 131   | 74.8%    | 84.6% | +9.8%    | 12.7%   | 2.8%     |
| 132   | 74.2%    | 84.6% | +10.3%   | 12.6%   | 2.8%     |
| 133   | 73.3%    | 84.6% | +11.3%   | 11.6%   | 2.8%     |
| 134   | 72.2%    | 84.5% | +12.3%   | 11.9%   | 2.8%     |
| 135   | 71.9%    | 84.4% | +12.5%   | 11.6%   | 2.8%     |
| 136   | 70.8%    | 84.3% | +13.5%   | 10.9%   | 2.6%     |
| 137   | 71.7%    | 84.3% | +12.6%   | 11.0%   | 2.7%     |
| 138   | 70.6%    | 84.2% | +13.7%   | 9.5%    | 2.6%     |
| 139   | 70.8%    | 84.2% | +13.4%   | 9.4%    | 2.6%     |
| 140   | 69.6%    | 84.2% | +14.6%   | 8.7%    | 2.6%     |
| 141   | 69.3%    | 84.1% | +14.8%   | 9.1%    | 2.6%     |
| 142   | 69.0%    | 83.9% | +15.0%   | 8.8%    | 2.7%     |
| 143   | 69.4%    | 83.8% | +14.4%   | 9.2%    | 2.6%     |
| 144   | 67.8%    | 83.7% | +15.9%   | 7.1%    | 2.6%     |
| 145   | 69.0%    | 83.6% | +14.6%   | 7.9%    | 2.7%     |
| 146   | 68.4%    | 83.5% | +15.0%   | 7.4%    | 2.6%     |

|     |       |       |        |      |      |
|-----|-------|-------|--------|------|------|
| 147 | 68.0% | 83.4% | +15.3% | 6.2% | 2.6% |
| 148 | 68.4% | 83.2% | +14.9% | 6.9% | 2.7% |
| 149 | 66.9% | 83.1% | +16.2% | 5.7% | 2.8% |
| 150 | 67.0% | 83.1% | +16.1% | 6.0% | 2.8% |
| 151 | 67.0% | 83.1% | +16.1% | 5.6% | 2.8% |
| 152 | 66.8% | 82.9% | +16.1% | 5.3% | 2.9% |
| 153 | 67.1% | 82.8% | +15.7% | 4.9% | 2.8% |
| 154 | 66.6% | 82.7% | +16.1% | 4.6% | 2.8% |
| 155 | 65.7% | 82.8% | +17.1% | 5.0% | 2.8% |
| 156 | 66.5% | 82.7% | +16.2% | 4.9% | 2.8% |
| 157 | 66.6% | 82.7% | +16.1% | 4.7% | 2.9% |
| 158 | 66.8% | 82.7% | +15.9% | 4.8% | 2.8% |
| 159 | 66.2% | 82.6% | +16.4% | 4.6% | 2.8% |
| 160 | 66.3% | 82.5% | +16.3% | 4.7% | 2.9% |
| 161 | 66.3% | 82.4% | +16.2% | 4.7% | 2.8% |
| 162 | 65.9% | 82.4% | +16.5% | 4.5% | 2.8% |
| 163 | 66.0% | 82.3% | +16.3% | 4.0% | 2.8% |
| 164 | 65.5% | 82.2% | +16.7% | 4.6% | 2.8% |
| 165 | 65.5% | 82.1% | +16.6% | 4.6% | 2.8% |
| 166 | 66.2% | 82.0% | +15.8% | 4.5% | 2.8% |
| 167 | 65.7% | 81.9% | +16.2% | 4.8% | 2.8% |
| 168 | 65.0% | 81.8% | +16.8% | 4.1% | 2.8% |
| 169 | 65.2% | 81.8% | +16.6% | 4.3% | 2.8% |
| 170 | 65.3% | 81.7% | +16.4% | 4.6% | 2.8% |
| 171 | 64.9% | 81.6% | +16.7% | 4.5% | 2.8% |
| 172 | 64.8% | 81.5% | +16.7% | 4.6% | 2.8% |
| 173 | 64.6% | 81.5% | +16.8% | 3.9% | 2.9% |
| 174 | 64.8% | 81.5% | +16.7% | 3.9% | 2.9% |
| 175 | 65.7% | 81.3% | +15.6% | 5.4% | 2.9% |
| 176 | 65.1% | 81.3% | +16.2% | 4.4% | 2.9% |
| 177 | 65.0% | 81.2% | +16.2% | 4.3% | 3.0% |
| 178 | 65.2% | 81.1% | +15.9% | 4.6% | 3.1% |
| 179 | 64.5% | 81.0% | +16.5% | 4.7% | 3.0% |
| 180 | 64.8% | 80.9% | +16.1% | 4.3% | 2.9% |
| 181 | 64.7% | 81.0% | +16.3% | 4.3% | 3.0% |
| 182 | 64.6% | 80.9% | +16.3% | 4.7% | 2.9% |
| 183 | 64.6% | 80.9% | +16.3% | 4.5% | 2.9% |
| 184 | 65.0% | 80.8% | +15.8% | 4.5% | 3.0% |
| 185 | 64.1% | 80.8% | +16.7% | 4.5% | 3.0% |
| 186 | 64.4% | 80.7% | +16.3% | 4.1% | 3.0% |
| 187 | 64.8% | 80.7% | +16.0% | 4.1% | 3.0% |
| 188 | 64.0% | 80.7% | +16.6% | 4.6% | 2.9% |
| 189 | 64.5% | 80.6% | +16.1% | 4.0% | 3.0% |
| 190 | 64.3% | 80.5% | +16.2% | 4.4% | 2.9% |
| 191 | 64.2% | 80.4% | +16.1% | 4.1% | 3.0% |
| 192 | 64.1% | 80.2% | +16.2% | 4.3% | 3.0% |
| 193 | 63.8% | 80.2% | +16.5% | 3.6% | 3.0% |
| 194 | 63.9% | 80.2% | +16.3% | 3.9% | 3.0% |
| 195 | 64.4% | 80.0% | +15.6% | 4.6% | 2.9% |
| 196 | 63.8% | 80.0% | +16.2% | 4.4% | 2.8% |
| 197 | 63.4% | 79.9% | +16.4% | 4.1% | 2.8% |
| 198 | 64.0% | 79.9% | +15.8% | 3.9% | 2.9% |
| 199 | 63.5% | 79.9% | +16.4% | 3.8% | 3.0% |
| 200 | 64.0% | 79.7% | +15.7% | 4.3% | 3.0% |
| 201 | 63.9% | 79.8% | +15.9% | 4.0% | 2.9% |
| 202 | 63.8% | 79.5% | +15.7% | 4.0% | 2.9% |
| 203 | 63.6% | 79.5% | +15.9% | 4.3% | 3.0% |
| 204 | 64.1% | 79.4% | +15.4% | 4.1% | 3.0% |
| 205 | 63.7% | 79.4% | +15.6% | 4.6% | 2.9% |
| 206 | 63.8% | 79.3% | +15.6% | 4.4% | 2.9% |
| 207 | 63.5% | 79.2% | +15.7% | 4.0% | 2.9% |
| 208 | 63.6% | 79.1% | +15.6% | 4.0% | 3.0% |
| 209 | 63.7% | 79.1% | +15.4% | 4.2% | 2.9% |

|     |       |       |        |      |      |
|-----|-------|-------|--------|------|------|
| 210 | 64.0% | 79.0% | +15.0% | 4.2% | 2.9% |
| 211 | 63.4% | 78.9% | +15.6% | 4.1% | 2.9% |
| 212 | 63.5% | 78.9% | +15.4% | 4.3% | 3.0% |
| 213 | 63.2% | 78.7% | +15.5% | 3.8% | 3.0% |
| 214 | 63.2% | 78.8% | +15.6% | 4.7% | 2.9% |
| 215 | 63.3% | 78.8% | +15.5% | 4.3% | 3.1% |
| 216 | 63.6% | 78.6% | +15.0% | 4.3% | 3.0% |
| 217 | 63.1% | 78.6% | +15.4% | 3.8% | 3.1% |
| 218 | 63.1% | 78.5% | +15.4% | 3.5% | 3.0% |
| 219 | 63.3% | 78.5% | +15.2% | 3.8% | 3.0% |
| 220 | 63.0% | 78.5% | +15.5% | 3.7% | 3.1% |
| 221 | 63.2% | 78.5% | +15.3% | 3.9% | 3.0% |
| 222 | 63.1% | 78.3% | +15.2% | 3.9% | 3.1% |
| 223 | 63.1% | 78.3% | +15.2% | 3.7% | 3.2% |
| 224 | 63.1% | 78.3% | +15.2% | 3.7% | 3.2% |
| 225 | 62.9% | 78.2% | +15.3% | 3.8% | 3.1% |
| 226 | 63.3% | 78.2% | +14.9% | 3.7% | 3.1% |
| 227 | 62.9% | 78.0% | +15.1% | 4.2% | 3.1% |
| 228 | 63.2% | 77.9% | +14.7% | 3.4% | 3.1% |
| 229 | 63.6% | 77.9% | +14.3% | 3.6% | 3.0% |
| 230 | 63.3% | 77.9% | +14.6% | 3.6% | 3.0% |
| 231 | 62.9% | 77.9% | +14.9% | 3.5% | 3.1% |
| 232 | 62.9% | 77.8% | +14.9% | 3.3% | 3.0% |
| 233 | 63.0% | 77.7% | +14.8% | 3.8% | 2.9% |
| 234 | 62.4% | 77.7% | +15.3% | 3.7% | 3.0% |
| 235 | 62.9% | 77.7% | +14.8% | 3.5% | 2.9% |
| 236 | 62.9% | 77.6% | +14.7% | 3.8% | 3.0% |
| 237 | 62.7% | 77.5% | +14.9% | 3.7% | 3.0% |
| 238 | 62.8% | 77.5% | +14.7% | 3.9% | 3.1% |
| 239 | 62.4% | 77.3% | +14.9% | 3.7% | 3.1% |
| 240 | 62.6% | 77.3% | +14.7% | 3.4% | 3.1% |
| 241 | 63.0% | 77.3% | +14.2% | 3.5% | 3.1% |
| 242 | 63.0% | 77.3% | +14.3% | 3.7% | 3.1% |
| 243 | 62.6% | 77.2% | +14.6% | 4.0% | 3.2% |
| 244 | 62.4% | 77.1% | +14.7% | 3.6% | 3.2% |
| 245 | 62.5% | 77.1% | +14.6% | 3.7% | 3.1% |
| 246 | 62.4% | 77.0% | +14.6% | 3.8% | 3.0% |
| 247 | 62.2% | 77.0% | +14.8% | 3.8% | 3.1% |
| 248 | 62.3% | 76.9% | +14.6% | 3.7% | 3.0% |
| 249 | 62.1% | 76.8% | +14.7% | 3.4% | 3.1% |
| 250 | 61.5% | 76.7% | +15.3% | 3.9% | 3.1% |
| 251 | 62.0% | 76.6% | +14.5% | 3.3% | 3.1% |
| 252 | 62.2% | 76.6% | +14.4% | 3.9% | 3.1% |
| 253 | 61.5% | 76.5% | +15.0% | 3.7% | 3.0% |
| 254 | 62.1% | 76.5% | +14.4% | 3.6% | 3.0% |
| 255 | 61.3% | 76.6% | +15.3% | 3.8% | 3.0% |
| 256 | 61.5% | 76.4% | +14.9% | 3.6% | 3.0% |
| 257 | 61.7% | 76.4% | +14.7% | 3.1% | 3.0% |
| 258 | 61.9% | 76.4% | +14.5% | 3.6% | 3.0% |
| 259 | 61.4% | 76.4% | +15.0% | 3.5% | 3.1% |
| 260 | 61.8% | 76.2% | +14.4% | 3.3% | 3.1% |
| 261 | 61.8% | 76.2% | +14.4% | 3.6% | 3.2% |
| 262 | 61.7% | 76.1% | +14.4% | 3.7% | 3.1% |
| 263 | 62.3% | 76.2% | +13.9% | 3.6% | 3.1% |
| 264 | 62.2% | 76.2% | +14.0% | 3.5% | 3.1% |
| 265 | 62.3% | 76.1% | +13.9% | 3.5% | 3.1% |
| 266 | 61.9% | 76.1% | +14.2% | 3.5% | 3.1% |
| 267 | 61.7% | 76.0% | +14.3% | 3.3% | 3.2% |
| 268 | 62.2% | 76.0% | +13.8% | 3.3% | 3.2% |
| 269 | 61.7% | 76.0% | +14.3% | 4.0% | 3.1% |
| 270 | 61.7% | 75.9% | +14.2% | 3.6% | 3.2% |
| 271 | 61.8% | 75.8% | +14.0% | 3.8% | 3.2% |
| 272 | 61.0% | 75.7% | +14.7% | 3.7% | 3.1% |

|     |       |       |        |      |      |
|-----|-------|-------|--------|------|------|
| 273 | 61.6% | 75.7% | +14.1% | 3.7% | 3.2% |
| 274 | 61.0% | 75.6% | +14.6% | 3.9% | 3.1% |
| 275 | 61.5% | 75.7% | +14.2% | 3.9% | 3.1% |
| 276 | 61.6% | 75.6% | +14.1% | 3.3% | 3.2% |
| 277 | 61.3% | 75.5% | +14.3% | 3.5% | 3.1% |
| 278 | 61.9% | 75.4% | +13.5% | 3.5% | 3.1% |
| 279 | 61.3% | 75.5% | +14.2% | 2.8% | 3.1% |
| 280 | 61.3% | 75.5% | +14.2% | 3.3% | 3.0% |
| 281 | 61.3% | 75.5% | +14.2% | 3.5% | 3.1% |
| 282 | 61.0% | 75.4% | +14.3% | 3.8% | 3.1% |
| 283 | 61.3% | 75.3% | +14.0% | 3.6% | 3.2% |
| 284 | 61.3% | 75.3% | +14.0% | 3.7% | 3.1% |
| 285 | 61.3% | 75.2% | +13.8% | 3.7% | 3.2% |
| 286 | 61.2% | 75.2% | +14.0% | 3.7% | 3.1% |
| 287 | 61.0% | 75.2% | +14.2% | 3.4% | 3.2% |
| 288 | 61.2% | 75.1% | +13.9% | 3.5% | 3.3% |
| 289 | 61.4% | 75.1% | +13.7% | 3.7% | 3.3% |
| 290 | 61.0% | 75.2% | +14.2% | 3.6% | 3.2% |
| 291 | 61.1% | 75.1% | +13.9% | 3.7% | 3.2% |
| 300 | 61.1% | 74.7% | +13.6% | 3.2% | 3.1% |
| 310 | 61.3% | 74.4% | +13.0% | 3.6% | 3.1% |
| 320 | 60.9% | 74.0% | +13.1% | 3.6% | 3.1% |
| 330 | 60.9% | 73.6% | +12.8% | 3.3% | 3.2% |
| 340 | 60.4% | 73.4% | +13.0% | 3.7% | 3.2% |
| 350 | 60.4% | 73.0% | +12.5% | 3.6% | 3.1% |

|                                                                                                                    |  |  |
|--------------------------------------------------------------------------------------------------------------------|--|--|
| Hopfield capacity: M=124 ( $\alpha=0.121$ )<br>CONN capacity: M=195 ( $\alpha=0.190$ )<br>Improvement ratio: 1.57× |  |  |
|--------------------------------------------------------------------------------------------------------------------|--|--|

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Scaling run completed.